🤌 XAI Lecture 11

Counterfactual Explanations & Algorithmic Recourse

FAMAF - UNC

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🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Paper 1 ---

COUNTERFACTUAL EXPLANATIONS WITHOUT OPENING THE BLACK BOX: AUTOMATED DECISIONS AND THE GDPR

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(Wachter, Mittelstadt, and Russell 2018)

Introduction + Motivation ---

EU General Data Protection Regulation (2018)

- "the toughest privacy and security law in the world"
- Article 13-14, regarding automated decision-making: "meaningful information about the logic involved"
- Recital 71: "the right to obtain an explanation of the decision reached and to challenge the decision"

4 Key Problems

1. Not legally binding
2. Only applicable in limited cases
3. Explainability is technically very challenging
4. Competing interests of data controllers, subjects, etc.

🧉 Unconditional Counterfactual Explanations ---

Authors propose

- **3 aims for explanations**
  - Inform and help the subject understand "why"
  - Provide grounds to contest decisions
  - Understand options for recourse
- **CFEs**
  - Fulfill the above goals
  - Overcome challenges w.r.t. current interpretability work
  - Can bridge the gap between interests of data subjects and data controllers

Do you buy this?

🧉 Unconditional Counterfactual Explanations ---

- **Definition:**
  - A statement of how the world would have to be different for a desirable outcome to occur
  - **Because** of features $\{x_1, x_2\}$, x's outcome was label y_1 **If** $x = \{x_1 + \delta_1, x_2 + \delta_2\}$, **then** y_2.
- **Example:**
  - "You were denied a loan **because** your annual income was \$30000. **If** your income has been \$45000, **you would have been offered a loan**"

Notes

- No one **unique** CF
- Need to consider **actionability** (mutability of variables)
- Providing **several diverse** CFEs may be more useful than simply the closest/shortest one

🧉 Background + Related Work ---

- Historic context of knowledge
- Prior explainability work
- Adversarial examples/perturbations
- Causality/Fairness

🧉 Background: Historic Context of Knowledge ---

- In order to know something, it is not enough to simply **believe** that it is true: rather, you must also have a good **reason** for believing it
- If q were false, S would not believe p

Note

- This statement only describes S's **beliefs**, which might not reflect reality
- This statement can be made without knowledge of the **causal relationship** between p and q
- Q: who is S? The user? The model? Us?

🧉 Background: Previous Explanations in AI/ML ---

- **Previously:** providing insight into the internal state of an algorithm, human-understandable approximations of the algorithm
- Three-way tradeoff between:
  1. quality of approximation
  2. ease of understanding the function
  3. size of the domain for which the approximation is valid
- **CFEs:**
  - Minimal amount of information
  - Require no understanding of internal logic of model
  - No approximation (although might not always be minimum length)
  - Con: May be overly restrictive

🧉 Background: Adversarial Perturbations (not the same) ---

- **Shared Mechanism:** Both find the minimal change (δ) to flip a prediction: $f(x+\delta) = y'$.
- **The Contrast:**
  - **Adversarial:** Invisible, targets pixels, intended to **deceive**.
  - **Counterfactual:** *Sparse*, targets features, intended to **inform**.
- **The "Manifold" Requirement:**
  - Mathematical changes must be **plausible** in the real world.
  - GDPR requires explanations to be "meaningful," not just possible.
- **Key Insight:** A CFE is an adversarial attack with a **human-centric purpose**.

🧉 Background: Causality and Fairness ---

- **Bias Detection:** CFEs act as an audit tool; they reveal if a decision is based on protected attributes (e.g., race, gender, age).
- **Counterfactual Fairness:** If the "shortest path" to a loan approval requires changing your race or gender, the model is demonstrably discriminatory.
- **Evidence of Disparate Treatment:** CFEs provide a "smoking gun" for auditors without needing to inspect the model's internal weights.
- **The Converse Question:** If a model is biased, will the CFE *always* change a protected attribute?
  - *Warning:* Bias often hides in "proxies" (e.g., ZIP code as a proxy for race).

🧉 Summary of Contributions ---

Core argument: explaining automated decisions does not require opening the black box.

Contribution	Key point
Complexity of ML explanations	Internal logic is too complex to convey to lay users
Counterfactual approach	Rooted in adversarial ML; no model internals required
GDPR alignment	CFEs satisfy transparency goals better than local approximations

Bottom line: *CFEs shift the focus from how the model works to what would need to change — actionable, model-agnostic, and legally compatible.*

🧉 Approach: Counterfactual Optimization ---

$$\arg\min_{x'}\max_{\lambda}\ \lambda(f_w(x') - y')^2 + d(x_i, x')$$

In plain terms: *find the closest point to x_i that flips the prediction to y'.*

Symbol	Meaning
x'	Counterfactual point (what we optimize)
λ	Weight balancing target vs. proximity
$\lambda(f_w(x') - y')^2$	Penalizes missing the target label
$d(x_i, x')$	Penalizes distance from factual point
$\max_\lambda$	Forces label flip to dominate over proximity

Procedure: alternate — optimize x', then increase λ until constraint is satisfied.

🧉 Approach: Distance Metrics ---

$$d(x, x') = \sum_{k \in F} \frac{\|x_{i,k} - x'_k\|_p}{N_k}$$

In plain terms: sum of per-feature differences, normalized so all features are comparable.

Term	Meaning
$x_{i,k} - x'_k$	Change in feature k
$\|\cdot\|_p$	ℓ_1 (sparse changes) or ℓ_2 (smooth changes)
N_k	Normalizing factor — makes features comparable

Two normalizing choices:

1. $N_k = \mathrm{MAD}_k = \mathrm{median}_j(|X_{j,k} - \mathrm{median}_l(X_{l,k})|)$
2. $N_k = \mathrm{std}_j(x_{j,k})$

🧉 Experimental Results: LSAT 1/3 ---

Setup: predict law school admission score from {GPA, LSAT, Race} using a black-box model.

Goal:

- For each below-average student, find the *minimal change* that brings their score to $y' = 0$ (population mean).
- *population mean*The minimum score to be considered an average candidate.

Key question:

Does the choice of distance metric produce fair, realistic, and actionable counterfactuals?

🧉 Experimental Results: LSAT ---

score	Original Data GPA	LSAT	Race	Unnormalised L2 Counterfactuals GPA	LSAT	Race	Counterfactual Hybrid GPA	LSAT	Race
0.17	3.1	39.0	0	3.0	39.0	0.3	1.5	38.4	0
0.54	3.7	48.0	0	3.5	47.9	0.9	-1.6	45.9	0
-0.77	3.3	28.0	1	3.5	28.1	-0.3	5.3	28.9	0
-0.83	2.4	28.5	1	2.6	28.6	-0.4	4.8	29.4	0
-0.57	2.7	18.3	0	2.9	18.4	-1.0	8.4	20.6	0

score	GPA	LSAT	Race	Normalised L2 GPA	LSAT	Race	GPA	LSAT	Race
0.17	3.1	39.0	0	3.0	37.0	0.2	3.0	34.0	0
0.54	3.7	48.0	0	3.5	39.5	0.4	3.5	33.1	0
-0.77	3.3	28.0	1	3.5	39.8	0.4	3.4	33.4	0
-0.83	2.4	28.5	1	2.7	37.4	0.2	2.6	35.7	0
-0.57	2.7	18.3	0	2.8	28.1	-0.4	2.9	34.1	0

score	Original Data GPA	LSAT	Race	Normalised L1 (MAD) Counterfactuals Continuous GPA	LSAT	Race	Counterfactual Hybrid GPA	LSAT	Race
0.17	3.1	39.0	0	3.1	35.0	0.1	3.1	34.0	0
0.54	3.7	48.0	0	3.7	33.5	0.0	3.7	32.4	0
-0.77	3.3	28.0	1	3.3	34.4	0.1	3.3	33.5	0
-0.83	2.4	28.5	1	2.4	39.3	0.2	2.4	35.8	0
-0.57	2.7	18.3	0	2.7	35.8	0.1	2.7	34.9	0

- Without normalization, the optimizer freely modifies Race (-1.0, 0.9), producing impossible counterfactuals.
- MAD normalization makes changing Race relatively costly, keeping it fixed across all CFEs.
- Distance metric choice is not a technical detail — it directly determines fairness and plausibility of recommendations.

Experimental Results: Discussion ---

1. **Two practical design decisions for realistic CFEs:**

Technique	Purpose
Clipping	Truncate CFE values to observed data range
Clamping	Force categorical variables to valid values

2. **We need reasobales counterfactual targets:**

Dataset	Target y'	Interpretation
LSAT	0	Reach population average score
PIMA	0.5	Cross from high to low diabetes risk

Note: *the choice of y' is not neutral — it encodes a normative judgment about what counts as a "desirable" outcome.*

🧉 Explanations and The GDPR ---

GDPR Recital 71 requires that automated decisions include: the right to obtain an explanation and to challenge the decision.

Requirement	CFE response
Intelligible explanation	Simple “if-then” statements
No black-box exposure	CFEs depend only on external facts, not model internals
No trade secret violation	No algorithm details disclosed
Individual-level insight	Tailored to each data subject’s situation

> **Bottom line:** *CFEs do not require opening the black box — they satisfy GDPR goals while protecting data controllers’ interests.*

🧉 Advantages of Counterfactual Explanations ---

- Bypass explaining the *internal workings* of complex machine learning system
- Simple to compute and convey
- Provide information that is both easily digestible and practically useful
  - Understanding the reasons for a decision
  - Contesting decisions
  - Altering future behaviour to receive a preferred outcome

Possible mechanism to meet the explicit requirements and background aims of the GDPR

🧉 Legal Information Gaps on Contesting Decisions ---

The GDPR leaves critical gaps for individuals seeking to contest automated decisions:

- **Art. 16**
  - Data subject has the right to correct inaccurate data used to make a decision, but does not need to be informed of which data the decision depended
- **Art. 22**
  - Data subjects do not need to be informed of their right *not* to be subject to an automated decision

CFEs close these gaps:

By revealing which features were most influential, they give individuals the information needed to meaningfully contest a decision.

🧉 Explanations to Alter Future Decisions ---

The third goal of explanations: not just to understand or contest, but to guide the individual toward a better outcome.

Source	Position
GDPR	Silent — does not explicitly require forward-looking explanations
Art. 29 Working Party	Recommends suggestions on how to improve outcomes
CFEs	Naturally satisfy this goal — show *what to change* and *by how much*

Key advantage: *CFEs can address the impact of changing multiple variables simultaneously, producing realistic and actionable recourse paths.*

Empirical Evidence: Yacoby et al. (2020) ---

"If it didn't happen, why would I change my decision?": How Judges Respond to Counterfactual Explanations for the Public Safety Assessment

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Setup: 8 U.S. state court judges evaluated criminal risk assessments accompanied by CFEs.

Example CFE shown to judges: *"If this defendant had stable employment, the system would have classified them as low risk."*

Finding: CFEs did not change judicial decisions because:

1. Misinterpretation: Judges see it as a claim about the defendant, not the model.
2. Irrelevance: Once understood, judges find it irrelevant to the case.

Implication: CFEs can be legally compliant and mathematically correct yet fail entirely in practice due to user misinterpretation.

🧉 Discussion: Limitations of CFEs (Wachter et al.) ---

Who do these explanations actually serve?

- **Developers:** debugging and model improvement
- **Data subjects:** recourse and contestation
- **Lawmakers:** accountability and compliance

Two sources of misinterpretation:

- **Causality:** CFEs describe model behavior, not real-world causal effects
- **Sparsity bias:** mathematically minimal changes may be unrealistic or unachievable

Takeaway: *CFEs tell you where to go, not how to get there — this gap motivates Karimi et al.'s causal recourse framework.*

Paper 2 ---

Algorithmic Recourse:
from Counterfactual Explanations to Interventions

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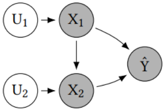
ABSTRACT

As machine learning is increasingly used to inform consequential decision-making (e.g., pre-trial bail and loan approval), it becomes important to explain how the system arrived at its decision, and also suggest actions to achieve a favorable decision. Counterfactual explanations –"how the world would have (had) to be different for a desirable outcome to occur"– aim to satisfy these criteria. Existing works have primarily focused on designing algorithms to obtain counterfactual explanations for a wide range of settings.



$$\left.\begin{aligned} X_1 &:= U_1 \\ X_2 &:= f_2(X_1) + U_2 \end{aligned}\right\} \mathcal{M}$$

$$\hat{Y} = h(X_1, X_2)$$

Figure 1: Illustration of an example causal generative process governing the world, showing both the graphical model, $\mathcal{G}$, and the structural causal model, $\mathcal{M}$, [34]. In this example,

(Karimi, Schölkopf, and Valera 2021)

🧉 Algorithmic Recourse ---

- Increasingly algorithms are used to make consequential decisions for individuals
- Recourse: "*Systematic process of reversing unfavorable decisions made by algorithms and bureaucracies*"
  - Promoting **agency** and **trust**

🧉 Paper's Main Contributions ---

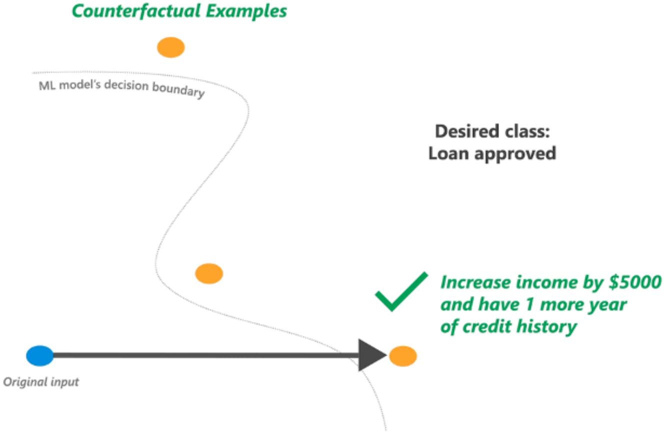
Starting point: CFEs tell you *where to go* but not *how to get there*.

- **Contribution 1 — Critique:** formalize the insufficiencies of CFE-based recourse, showing it fails when features are causally dependent.
- **Contribution 2 — Framework:** propose Recourse through Minimal Interventions (MINT), grounding recourse in Structural Causal Models (SCMs).
- **Contribution 3 — Validation:** demonstrate MINT on synthetic and real-world settings, achieving up to 65% cost reduction over CFE-based approaches.

Key shift:

From finding the nearest counterfactual *instance* to finding the minimal cost *action set* that achieves recourse.

🧉 Nearest Counterfactual Explanations ---



- For a person with features $\mathbf{x}^F$ who was denied a loan, find the "nearest neighbor" $\mathbf{x}$ who was granted a loan
- Difference between $\mathbf{x}^F$ and $\mathbf{x}$ is an "explanation" for the loan denial for $\mathbf{x}^F$

🧉 Nearest Counterfactual Explanation (CFE) ---

$$\mathbf{x}^{*\mathrm{CFE}} \in \arg\min_{\mathbf{x}} \mathrm{dist}(\mathbf{x}, \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}) \neq h(\mathbf{x}^F),\ \mathbf{x} \in \mathcal{P}$$

- **Finding nearest counterfactual explanation as an optimization problem**
  - $\mathbf{x}^{*\mathrm{CFE}}$ is the Counter Factual Explanation
  - Function h is the classifier
- **Distance metrics**
  - Lp norm; L1 norm divided by median absolute deviation; etc.

🧉 Why Are CFEs Not Ideal for Recourse? ---

CFEs provide **understanding** but do not necessarily lead to optimal **action recommendations**.

- **Problem 1 — Cost of actions (Ustun et al. 2019):**
  - CFEs assume all feature changes have equal and constant cost.
  - Example: going from \$0 to \$100k salary costs the same as \$800k to \$900k.
- **Problem 2 — Causal blindness (Karimi et al. 2021):**
  - CFEs ignore downstream effects of actions on causally dependent features.
  - Example: increasing salary by 14% automatically raises bank balance by 30% of that amount — CFEs miss this and recommend a costlier intervention.

Bottom line:

Acting on a CFE in the real world may be suboptimal, infeasible, or simply wrong.

🧉 Accounting for the "Cost" of Actions (Ustun et al. 2019) ---

Key insight: not all feature changes are equally easy — recourse should minimize *effort*, not just *distance*.

$$\begin{aligned} \delta^* \in \arg\min_{\delta} \ & \text{cost}(\delta; \mathbf{x}^F) \\ \text{s.t.} \quad & h(\mathbf{x}^{\text{CFE}}) \neq h(\mathbf{x}^F) \\ & \mathbf{x}^{\text{CFE}} = \mathbf{x}^F + \delta \\ & \mathbf{x}^{\text{CFE}} \in \mathcal{P}, \quad \delta \in \mathcal{F} \end{aligned}$$

In plain English:

Find the cheapest set of changes δ to apply to $\mathbf{x}^F$ such that the prediction flips, the result is plausible ($\mathcal{P}$), and the actions are feasible ($\mathcal{F}$) — e.g., cannot reduce age or change race. (Ustun, Spangher, and Liu 2019)

🧉 Insufficiencies of Ustun et al. Formulation ---

$$
\begin{aligned}
\delta^* \in \arg\min_{\delta} \; & \mathrm{cost}(\delta; \mathbf{x}^F) \\
\text{s.t.} \quad & h(\mathbf{x}^{\mathrm{CFE}}) \neq h(\mathbf{x}^F) \\
& \mathbf{x}^{\mathrm{CFE}} = \mathbf{x}^F + \delta \\
& \mathbf{x}^{\mathrm{CFE}} \in \mathcal{P}, \quad \delta \in \mathcal{F}
\end{aligned}
$$

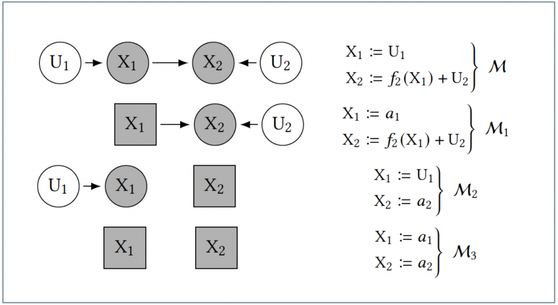
1. **Marginal cost of changing a feature is constant**
   - **Example:** Cost of going from salary \$0 to \$100k equals cost to go from salary \$800k to \$900k
2. **Doesn't consider downstream "causal" impact of actions**
   - **Example:** Loan decision for individual changed if "salary" reaches 100k (+33%) or "bank balance" reaches 30k (+20%). 20% change seems easier.
   - However, best **action recommendation** is to increase salary by 14% when 30% of salary automatically saved to bank.

🧉 Actions as Interventions: Structural Causal Model (SCM) 1/2 ---

Setup

- M ($M \in \Pi$) $= \langle F, X, U \rangle$: SCM
- X: endogenous (observed) variables
- U: exogenous (unobserved) variables
- F: $U \to X$, structural equations
- A ($\Pi \to \Pi$): structural interventions, i.e. transformations between SCMs
  - of the form $A := \mathrm{do}(\{X_i := a_i\}_{i \in I})$

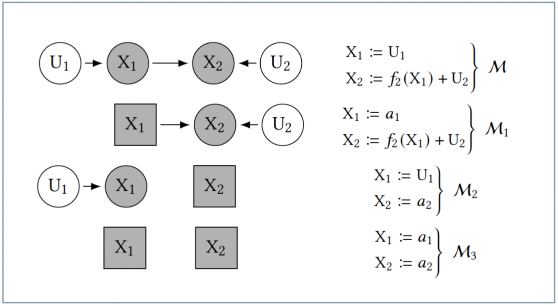
Structural Interventions



Actions as Interventions: Structural Causal Model (SCM) 2/2 ---

- $\mathcal{M}$ **(real world):** X_2 depends on X_1 — causal edge exists.
- $\mathcal{M}_1$ **(intervene on X_1):** set $X_1 = a_1$, edge remains — changing salary drags bank balance along.
- $\mathcal{M}_2$ **(intervene on X_2):** set $X_2 = a_2$, edge is severed — bank balance forced independently of salary.
- $\mathcal{M}_3$ **(intervene on both):** set $X_1 = a_1$, $X_2 = a_2$ — fully independent world. This is what CFEs assume, hence suboptimal.

Structural Interventions



The key observation

$\mathcal{M}_1 \neq \mathcal{M}_2 \neq \mathcal{M}_3$ — each intervention produces a different post-intervention world with different consequences.

🧉 Actions as Interventions: Structural Counterfactuals ---

If individual x^F performs action set A in world $\mathcal{M}$, what will their new feature vector be?

Assumptions: - No hidden confounders (true SCM known) - F is invertible

Two-step procedure:

1. **Abduction** — $F^{-1}(x^F)$: Infer exogenous variables U from observed features
2. **Prediction** — $F_A(U)$: Propagate action A through the causal graph

$$x^{\text{SCF}} = F_A(F^{-1}(x^F))$$

In plain English

First figure out *who this person is* (their latent U), then compute *what happens to them* after the action, respecting all causal dependencies.

🧉 Limitations of CFE-Based Recourse: Formalism ---

Setup:

- x^F: factual individual (current features)
- δ^*: action recommendation from Ustun et al.
- $I = \{i \mid \delta_i^* \neq 0\}$: set of features to be changed

Definition (CFE-based actions): $A^{\text{CFE}} := \text{do}(\{X_i := x_i^F + \delta_i^*\}_{i \in I})$

Proposition: A^{CFE} guarantees recourse **if and only if** the descendants of I in G are empty.

Corollary: if all features are root-nodes of G (independent world), CFE-based actions always guarantee recourse.

🧉 Limitations of CFE-Based Recourse: Formalism In plain English ---

- **Definition:** force each feature in I to its counterfactual value.
- **Proposition:** CFEs only guarantee recourse if changing X_i does not drag any other variable along.
- **Corollary:** CFEs always work in a fully independent world — exactly the world they implicitly assume, and one that rarely exists in practice

🧉 Algorithmic Recourse via Minimal Interventions: Setup ---

$$
\begin{aligned}
A^* \in \arg\min_{A} \ & \mathrm{cost}(A; \mathbf{x}^F) \\
\text{s.t.} \quad & h(\mathbf{x}^{\mathrm{SCF}}) \neq h(\mathbf{x}^F) \\
& \mathbf{x}^{\mathrm{SCF}} = \mathbb{F}_A(\mathbb{F}^{-1}(\mathbf{x}^F)) \\
& \mathbf{x}^{\mathrm{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}
\end{aligned}
$$

Remarks:

- $A^* \in \mathcal{F}$: optimal action set within feasible interventions
- $\mathrm{cost}(\cdot\,; \mathbf{x}^F) : \mathcal{F} \times X \to \mathbb{R}_+$: user-specified action cost
- $\mathbf{x}^{\mathrm{SCF}} = \mathbb{F}_A(\mathbb{F}^{-1}(\mathbf{x}^F))$: structural counterfactual respecting causal dependencies
- $\mathbf{x}^{\mathrm{SCF}} \neq \mathbf{x}^{\mathrm{CFE}}$: the resulting instance differs from Ustun et al.'s solution

In plain English

No longer ask *"what point should I reach?"* but *"what actions should I take?"* — accounting for the fact that actions propagate through the causal structure of the world.

🧉 Algorithmic Recourse via Minimal Interventions: Formalism ---

$$
\begin{aligned}
A^* \in \underset{A}{\arg\min}\ & \mathrm{cost}(A; \mathbf{x}^F) \\
\text{s.t.} \quad & h(\mathbf{x}^{\mathrm{SCF}}) \neq h(\mathbf{x}^F) \\
& \mathbf{x}^{\mathrm{SCF}} = F_A(F^{-1}(\mathbf{x}^F)) \\
& \mathbf{x}^{\mathrm{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}
\end{aligned}
$$

Proposition:

- A^{CFE}: recourse action derived from nearest counterfactual explanation
- A^*: recourse action from MINT

$$\mathrm{cost}(A^*; \mathbf{x}^F) \leq \mathrm{cost}(A^{\mathrm{CFE}}; \mathbf{x}^F)$$

In plain English

MINT always finds a recourse action that is at least as cheap as the CFE-based recommendation — and strictly cheaper whenever features are causally dependent.

🧉 Algorithmic Recourse via Minimal Interventions: MINT ---

Core requirement: ability to compute $x^{\text{SCF}} = F_A(F^{-1}(x^F))$ for *any* feasible action $A \in \mathcal{F}$.

Tractability assumption: SCM is an additive noise model (ANM):

$$X_i := f_i(\text{pa}_i) + U_i$$

Solution: Abduction-Action-Prediction (Pearl et al.):

- **Abduction:** infer U from observed x^F via $F^{-1}(x^F)$
- **Action:** modify SCM according to $\text{do}(\{X_i := a_i\}_{i \in I})$
- **Prediction:** propagate through modified SCM to obtain x^{SCF}

In plain English

ANMs make F invertible, enabling exact computation of the structural counterfactual for any action — which is what MINT needs to solve the optimization.

🧉 Abduction-Action-Prediction to Obtain x^{SCF} ---

$\{U_i\}_{i=1}^4$: mutually independent, exogenous
$\{f_i\}_{i=1}^4$: structural equations
$x = [x_1^F, x_2^F, x_3^F, x_4^F]^T$: observed factual features

1. **Abduction:** compute exogenous variables
   - $u_1 = x_1^F$, $u_2 = x_2^F$, $u_3 = x_3^F - f_3(x_1^F, x_2^F)$, $u_4 = x_4^F - f_4(x_3^F)$
2. **Action:** modify SCM with interventions
   - $X_1 := [1 \in I] \cdot a_1 + [1 \notin I] \cdot U_1$
   - $X_2 := [2 \in I] \cdot a_2 + [2 \notin I] \cdot U_2$
3. **Prediction:** recursively compute endogenous variables
   - $x_1^{\text{SCF}} := [1 \in I] \cdot a_1 + [1 \notin I] \cdot u_1$
   - $x_3^{\text{SCF}} := [3 \in I] \cdot a_3 + [3 \notin I] \cdot (f_3(x_1^{\text{SCF}}, x_2^{\text{SCF}}) + u_3)$

🧉 General Formulation and Solving the Optimization Problem ---

General Formulation

$$A^* \in \arg\min_{A} \text{cost}(A; \mathbf{x}^F)$$

$$\text{s.t.} \quad h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F)$$

$$x_i^{\text{SCF}} = [i \in I] \cdot (x_i^F + \delta_i)$$

$$+[i \notin I] \cdot (x_i^F + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^F))$$

$$\mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}$$

Remarks

- $x_i^F + \delta_i$: intervention
- $f_i(\mathbf{pa}_i^F)$: factual values of x_i's parents
- $f_i(\mathbf{pa}_i^{\text{SCF}})$: counterfactual values of x_i's parents
- new closed-form expression for $F_{A^*}(F^{-1}(x^F)) \rightarrow$ use optimization methods

🧉 Experimental Setup ---

Synthetic setting:
Generate data following causal generative process

Real-world setting:
Use existing German credit dataset to learn structural causal model equations, by fitting a linear regression

Cost for both is ℓ_1 norm over normalized feature change

🧉 Experimental Setup: Synthetic Setting ---

Causal Generative Process

$$U_1 \sim \$10000 \cdot \text{Poisson}(10),\ U_2 \sim \$2500 \cdot N(0,1)$$

$$X_1 := U_1$$

$$X_2 := f_2(X_1) + U_2, \quad X_2 := \frac{3}{10} \cdot X_1 + U_2$$

$$\hat{Y} = h(X_1, X_2), \quad h = \text{sgn}(X_1 + 5 \cdot X_2 - \$225000)$$

🧉 Experimental Results: Synthetic Setting ---

[annual salary, bank balance]

🧉 Experimental Results: Synthetic Setting (cont.) ---

$\mathbf{x}^{*\mathrm{SCF}}$ further dist from $\mathbf{x}^{F}$ than $\mathbf{x}^{*\mathrm{CFE}}$

BUT

$\mathrm{cost}(\delta^{*}; x^{F}) \approx 2\,\mathrm{cost}(A^{*}; x^{F})$

🧉 Experimental Setup: Real-World Setting ---

Structural Causal Model

$$X_1 := U_1, \quad X_2 := U_2$$

$$X_3 := f_3(X_1, X_2) + U_3$$

$$X_4 := f_4(X_3) + U_4$$

$$\hat{Y} = h(\{X_i\}_{i=1}^{4})$$

h can be logistic regression or decision tree

🧉 Experimental Results: Real-World Setting ---

[gender, age, credit given, credit repayment duration]

🧉 Experimental Results: Real-World Setting (cont.) ---

42% decrease in cost using Karimi et al.'s formulation

Averaged over 50 test individuals, $39 \pm 24\%$ and $65 \pm 8\%$ decrease in cost, for h as logistic regression and decision tree, respectively

🧉 Future Work: Extended Kinds of Interventions ---

Forms

- **Structural/hard** (actions in Karimi et al.): unconditionally sever all edges incident on intervened node
- **Additive/soft:** do not sever incident edges

$$x_i^{\text{SCF}} = [i \in I] \cdot \delta_i + (x_i^F + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^F))$$

Scopes

- Karimi et al. assumes action = intervention on endogenous variable
- **Fat-hand/non-atomic:** confounded/correlated interventions

Feasibility

Can encode as constraints to amend to $A \in \mathcal{F}$

- **Immutable:** closed under ancestral relationships
- **Mutable but non-actionable:** $[i \notin I] = 1$ is sufficient
- **Actionable and mutable:** contingent on (a) pre-intervention value of variable (b) pre-intervention value of other variables (c) post-intervention value of variable (d) post-intervention value of other variables

Future Work: Current Limitations ---

- **Reliance on true causal model of the world**

Concluding Thoughts ---

- Overall an interesting work bridging counterfactual explanations (previous papers) and algorithmic recourse (next papers), clarifying their differences
- Tradeoff between generalizability of setting and hardness of optimization problem
  - Ustun et al. pursue algorithmic recourse in a specific linear setting
  - Karimi et al. pursue algorithmic recourse in general settings, require true causal model of world
  - Open question: where do we stand on this tradeoff? Consider the origins of counterfactual explanations in Wachter et al.

Thank You!

References --- |

Karimi, Amir-Hossein, Bernhard Schölkopf, and Isabel Valera. 2021. "Algorithmic Recourse: From Counterfactual Explanations to Interventions," 353–62.

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